

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION

SUMMER SEMESTER, 2021-2022

DURATION: 1 HOUR 30 MINUTES

FULL MARKS: 50

CSE 4407: System Analysis and Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) Define the seven phases of the systems development life cycle (SDLC). 7
(CO1)
(PO1)
- b) Define tangible and intangible costs and benefits with examples. 4
(CO1)
(PO1)
- c) Define technical, economic, and operational feasibility. 6
(CO1)
(PO1)
2. a) The music platform, **gtunes** from **Mapple**, is undergoing a migration process from its outdated technology to a new, innovative solution. Table 1 shows the revenue (in \$USD) expected for the first four years of operation, in addition to the costs (in \$USD) for development, operation, and so on. 9
(CO1)
(PO1)
Using cash-flow analysis, determine the year when **Mapple** is expected to show a profit.

Table 1: Cash flow of the new system by year for Question 2 (a)

	Year 1	Year 2	Year 3	Year 4
Revenues				
Sales from music download	-	70,000	1,00,000	2,00,000
Customer subscription	-	1,50,000	2,00,000	2,50,000
Revenues from ads	-	1,50,000	2,00,000	3,00,000
Costs				
Development Costs				
Analysis and Design	42,000			
Development	1,20,000			
Software	25,000			
Hardware	50,000			
Operational Cost				
Employee Salary	2,00,000	2,50,000	2,75,000	2,90,000
Software License		15,000	15,000	15,000
Software Upgrades		15,000	5,000	5,000
Hardware Upgrades		20,000	5,000	5,000
Promotions	10,000	5,000	5,000	5,000

- b) Following **gtunes**, another popular music platform **Spotify** wants to upgrade their system to a newer technology. The present system's cost and the proposed system's cost (in \$USD) of **Spotify** is given in Table 2: 8
(CO1)
(PO1)
Using break-even analysis, determine the year in which **Spotify** will break-even.

Table 2: Proposed System costs and Present System costs by year for Question 2 (b)

	Proposed System Cost	Present System Cost
Year 1	84,000	64,500
Year 2	69,400	68,800
Year 3	60,900	74,100
Year 4	64,500	80,500

3. a) Cherry Jones owns a homeopathic medicine company called **Faithhealers**. She sells vitamins and other relatively nonperishable products for those who want choices regarding alternative medicine. Cherry is developing a new system that would require her staff to be retrained. Given the information in Table 3, draw a **GANTT** chart and identify the task that will take the most time to finish with proper justification.

6
(CO2)
(PO1)

Table 3: Dependency table of tasks for Question 3 (a, b, c)

Task Description	Task	Must Follow (Predecessor)	Time (Weeks)
Interview executives	A	None	6
Interview staff in order fulfillment	B	None	3
Design input prototype	C	B	2
Design output prototype	D	A, C	3
Write use cases	E	A, C	4
Record staff reactions to prototypes	F	D	2
Develop system	G	E, F	5
Write up training manual	H	B, G	3
Train staff working in order fulfillment	I	H	2

- b) Given the information in Table 3, draw a **Pert Diagram** and indicate the **critical path** that will take the most time with proper justification.
- c) Given the information in Table 3, prepare a **Critical Path Method (CPM) Diagram** and indicate the **critical path** that will take the most time with proper justification.

6
(CO2)
(PO1)

4
(CO2)
(PO1)

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DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2021-2022
FULL MARKS: 75

HUM 4441: Engineering Ethics

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. Michael Scott is a software engineer at Dunder Mifflin. His main job is developing softwares. One day, while his colleague Creed Bratton (Quality Assurance Engineer) was out sick, Michael finalised a module without any confirmation from Creed. The software turned out to be slowing down devices. There was chaos within the company. Dwight Schrute, who is Assistant to the regional manager found out that it was Michael's fault and he told the manager. When accused, Michael denied his mistakes.
 - a) Analyse the mistakes Michael made concerning the NSPE codes and identify the codes he breached. 10
(CO3)
(PO8)
 - b) Classify the type of whistle-blowing that Dwight engage in. 4
(CO2)
(PO7)
 - c) Evaluate the necessity of whistle-blowing for Dwight with the four conditions required. 6
(CO2)
(PO7)
 - d) Explain the role of the manager in preventing whistle-blowing in the first place. 5
(CO2)
(PO7)
2. a) The case of "The City of Albuquerque vs. Isleta Pueblo" involves the Isleta Pueblo, a well-known sovereign Indian tribe located in Albuquerque, New Mexico. The Pueblo is concerned about the high concentration of arsenic in the Rio Grande, which they use for drinking during religious ceremonies. They seek to enforce a stricter water quality standard, which is twice as stringent as the EPA regulations, for the water discharged by Albuquerque. The EPA sides with Pueblo, citing federal law giving Indian reservations the right to set their own pollution standards. The city of Albuquerque argues that Pueblo's standards are too strict, unnecessary, and a violation of the First Amendment's prohibition of government-established religion. The case was taken to the U.S. Supreme Court, but the court turned down Albuquerque's appeal and allowed the new EPA arsenic standards based on Isleta's requirements to stand. The city and other similarly affected municipalities are seeking federal aid to meet these new standards, which are of interest to many other states and municipalities in the west.

Show the perspectives of utilitarianism, rights and duty ethics, and virtue ethics about the case.

9
(CO1)
(PO6)
- b) Is there an ethical obligation to ensure that the information you post on your Internet web-site is accurate and true; or, is it up to the Web user to be discriminating and to realize that some material might not be accurate? Present your argument based on ethical theories on the usage of computers. 8
(CO1)
(PO6)

c) Should computer and software designers be concerned about the possible abuse of their products? Should designs incorporate methods for preventing the misuse of computers? Evaluate with proper justification. 8 (CO2) (PO7)

3. a) Late in 1994, reports began to appear that the latest generation of Pentium microprocessors was flawed. The stories reported that computers equipped with these chips were unable to correctly perform some relatively simple multiplication and division operations. At first, Intel, the manufacturer of the Pentium microprocessor, denied that there was a problem. Later, it argued that although there was a problem, the error would be significant only in sophisticated applications, and most people wouldn't even notice that an error had occurred. It was also reported that Intel had been aware of the problem and already was working to fix it. As a result of this publicity, many people who had purchased Pentium-based computers asked to have the defective chip replaced. Until the public outcry had reached huge proportions (The New York Times), Intel refused to replace the chips. Finally, when it was clear that this situation was a public relations disaster for them, Intel agreed to replace the defective chips when customers requested it. 20 (CO3) (PO8)

Analyze the situation using the Line Drawing Technique to find out if what Intel did was ethical or not. Use the following cases to populate the line:

- There is a flaw in the chip, but it truly is undetectable and won't affect any customer's applications.
- There are flaws in the chip, the customer is informed of them, but no help is offered.
- A warning label says that the chip should not be used for certain applications.
- Recall notices are sent out, and all flawed chips are replaced.
- All allegations are denied and the company refuses to replace anything as it has already been bought.
- Replacement chips are offered only if the customer notices the problem.

b) A company sales representative would like you to attend a one-day sales seminar in Dhanmondi. Your company will pay for your trip. Is this considered a bribe? How about if the meeting is in Switzerland? What if the sales representative's company is going to cover your travel expanses? What if your family is invited as well? 5 (CO2) (PO7)

Compare all the cases and identify the bribes and gifts.

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SUMMER SEMESTER, 2021-2022
FULL MARKS: 75

CSE 4403: Algorithms

Programmable calculators are not allowed. Do not write anything on the question paper.
 Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) Describe various notations to estimate the growth of a function. Using the big O notation, find the runtime complexity of the code snippets provided below (consider all possible value of N and M): 6+6
(CO1)
(PO1)

i.

```
1. for (int i = 0; i < N; i++)
2.     cout << "Hello";
3. for (int i = 0; i < M; i++)
4.     cout << "Hello";
```

Code Snippet 1: Code Snippet for question 1.a)

ii.

```
1. for (int i = 0; i < N; i++)
2.     for (int j = 1; j < M; j++)
3.         j *= 2;
```

Code Snippet 2: Code Snippet for question 1.a)

iii.

```
1. for (int i = 0; i < N; i++)
2.     for (int j = 1; j < M; j *= 2)
3.         for (int k = 1; k < 100; k++)
4.             cout << "Hello";
```

Code Snippet 3: Code Snippet for question 1.a)

- b) Imagine you are playing the classic game of snake, where you can navigate the snake to left, right, up, and down. You are given a 2D map containing dot (.) and exclamation mark (!). You can start the game by placing the snake at any location with dot. The snake can move along the dots (.) only, it cannot go through an exclamation mark (!). The snake is allowed to visit one dot more than once. Your task is to find how many snakes will be needed to visit all the dots of the map.

- i. Design a suitable algorithm to solve the above-mentioned problem.

10
(CO3)
(PO3)

- ii. Formulate the worst runtime complexity of your proposed solution.

3
(CO2)
(PO2)

5 W=V
N=N
NCC V.

2. Imagine you appeared for a mid-semester exam in IUT. You have a total time of V minutes to answer as many questions as you can from a set of N number of questions. Each question requires a specific amount of time to answer and rewards a certain amount of marks. No mark will be awarded for a partial answer to a question. For the i^{th} question ($1 \leq i \leq N$), you will be provided with the time t_i and mark m_i .

Your goal is to find the maximum obtainable marks in the mid-semester exam.

- a) Explain why this problem can be solved using dynamic programming (DP).

5
(CO1)
(PO1)

- b) Design a bottom-up algorithm for solving this problem and find the initial values for the DP table.

10
(CO3)
(PO3)

OR

Design a recursive memoized algorithm for this problem and find the best cases.

- c) Formulate the worst-case time complexity for your proposed algorithm.

3
(CO2)
(PO2)

- d) Construct a DP table for the following configuration of the above-mentioned problem:

$V = 8$ minutes Time required for each question: {3, 4, 6, 5}
 $N = 4$ Mark for each question: {2, 3, 1, 4}

7
(CO2)
(PO2)

3. a) There are N cities in a country numbered as 1, 2, 3, ..., N . The capital of the country is numbered as 0. There are M one-way walking paths in the country connecting two cities. To encourage the people of the country to walk more, the government announced financial rewards when someone walks from one city to another. In contrast, someone must pay a fixed amount of money as a tax whenever s/he reaches a city by walking. Suppose, you start your walking from a given city and always end your journey at the capital city. You will be provided with the reward amount of M paths each of which directly connects a pair of cities and the tax amount for all N cities. Given your starting city, your goal is to find the path to the capital city to earn the highest reward.

- i. Analyze the above-mentioned problem and design a suitable algorithm to solve it.

8
(CO3)
(PO3)

- ii. Justify the reason for choosing your solution.

4
(CO2)
(PO2)

- iii. Formulate the worst time complexity of your provided solution.

3
(CO2)
(PO2)

- b) i. Find a solution for a modified variant of the problem described in question 3.a), where your goal is to find the most suitable starting city so that you can maximize your reward.

8+2
(CO3)
(PO3)

- ii. Will there be any scenario, where this problem will be unsolvable? Justify your answer.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Mid-Semester Examination
Course No.: EEE 4483
Course Title: Digital Electronics and Pulse Techniques

Summer Semester, A. Y. 2021-2022
Time: 90 Minutes
Full Marks: 75

There are 3 (three) questions. Answer all 3 (three) questions. The symbols have their usual meanings. Programmable calculators are not allowed. Marks of each question and corresponding COs and POs are written in the brackets. Do not write on the question paper.

1. a) Briefly describe the current hogging phenomenon in DCTL gates.

[5]
(CO1,
PO1)

- b) In Fig. 1, gate G1 is a 3-input DTL NAND gate that has a fan-out equal to N (output of G1 is driving one of the inputs of N other similar 3-input NAND gates). If all the inputs of gate G1 is at logical HIGH state, determine the approximate value of N. [Here, $\beta = 50$ and $\sigma = 0.85$]

[10]
(CO2,
PO2)

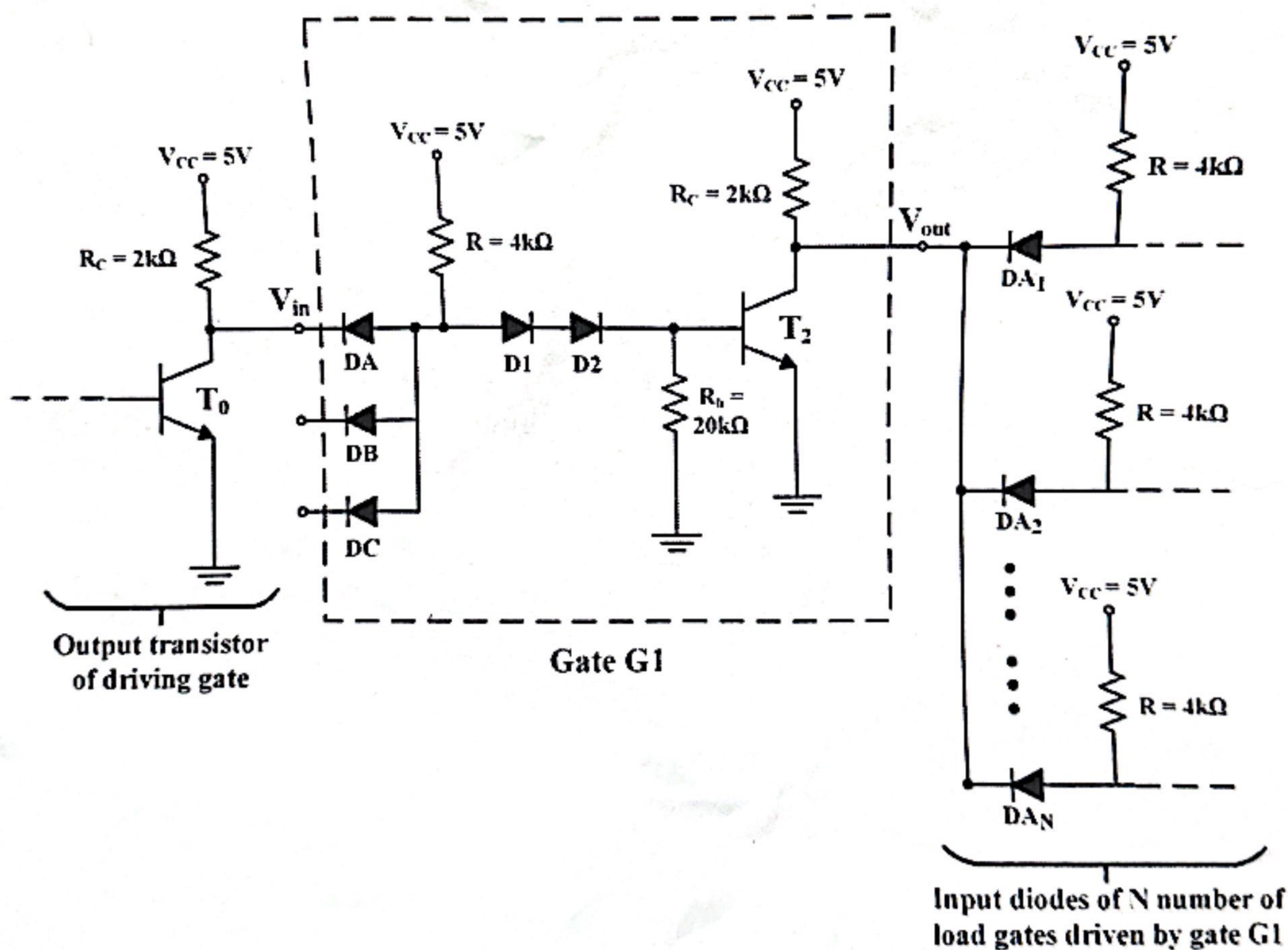


Fig. 1

- c) Sketch the circuit diagram of a 3-input TTL NAND gate with active pull-up. Specify the role of each transistor in the circuit and describe the overall operation of the circuit.
2. a) With proper reasoning, describe how TTL gates overcome the speed limitation of DTL gates.

[10]
(CO1,
PO1)

[12]
(CO1,
PO1)

- b) Describe which type of logic circuit is presented in Fig. 2. For this circuit, sketch the input-output characteristics curve, determine the different input & output voltage levels and noise margins. [Here, $\beta = 50$, $\sigma = 0.85$, $V_{BE}(\text{Cut-in}) = 0.65\text{V}$, $V_{BE}(\text{Sat}) = 0.7\text{V}$, $V_{CE}(\text{Sat}) = 0.2\text{V}$] [13] (CO2, PO2)

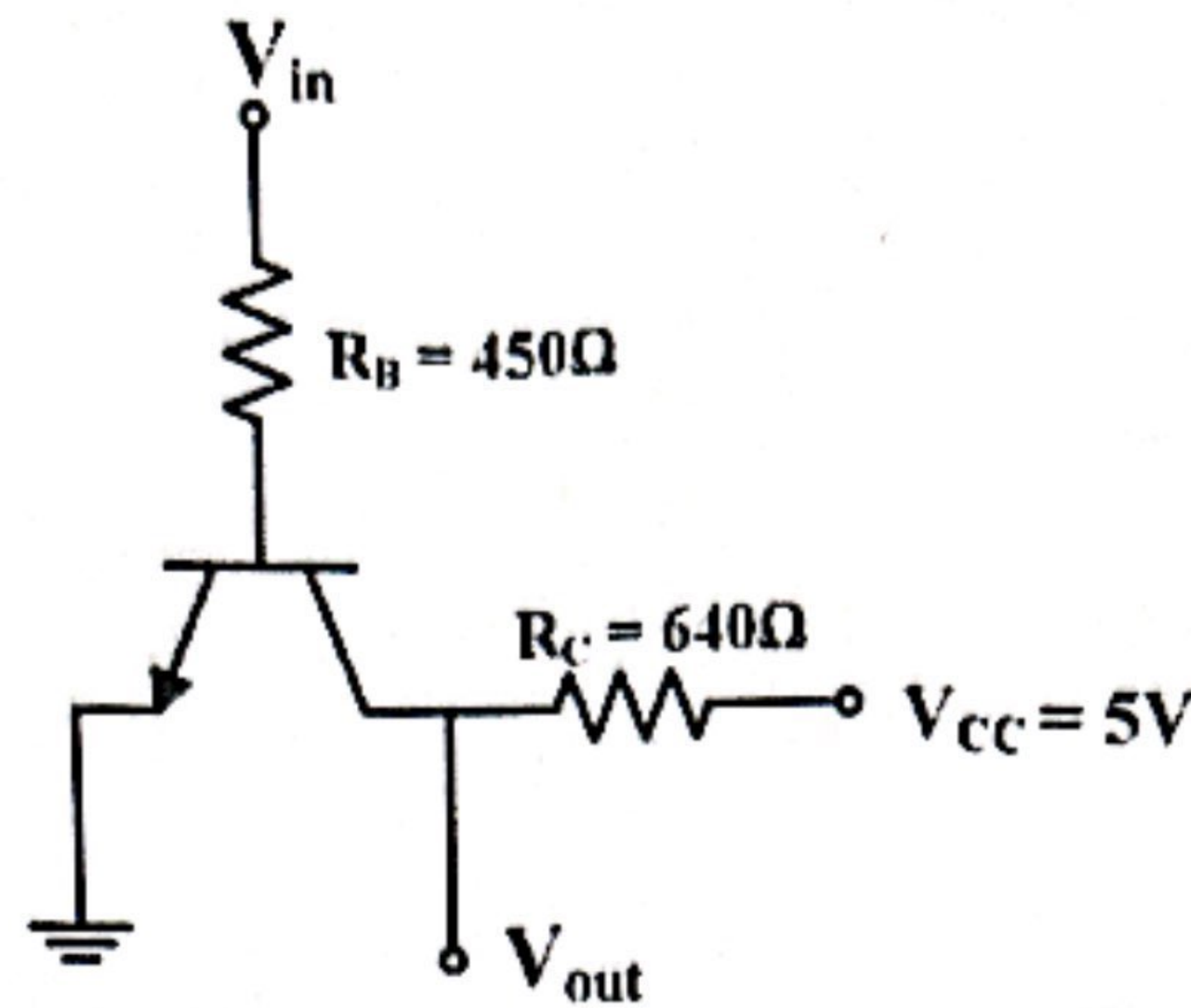


Fig. 2

3. a) Sketch the circuit diagram of the static CMOS design of the following circuit shown in Fig. 3. [10] (CO2, PO2)

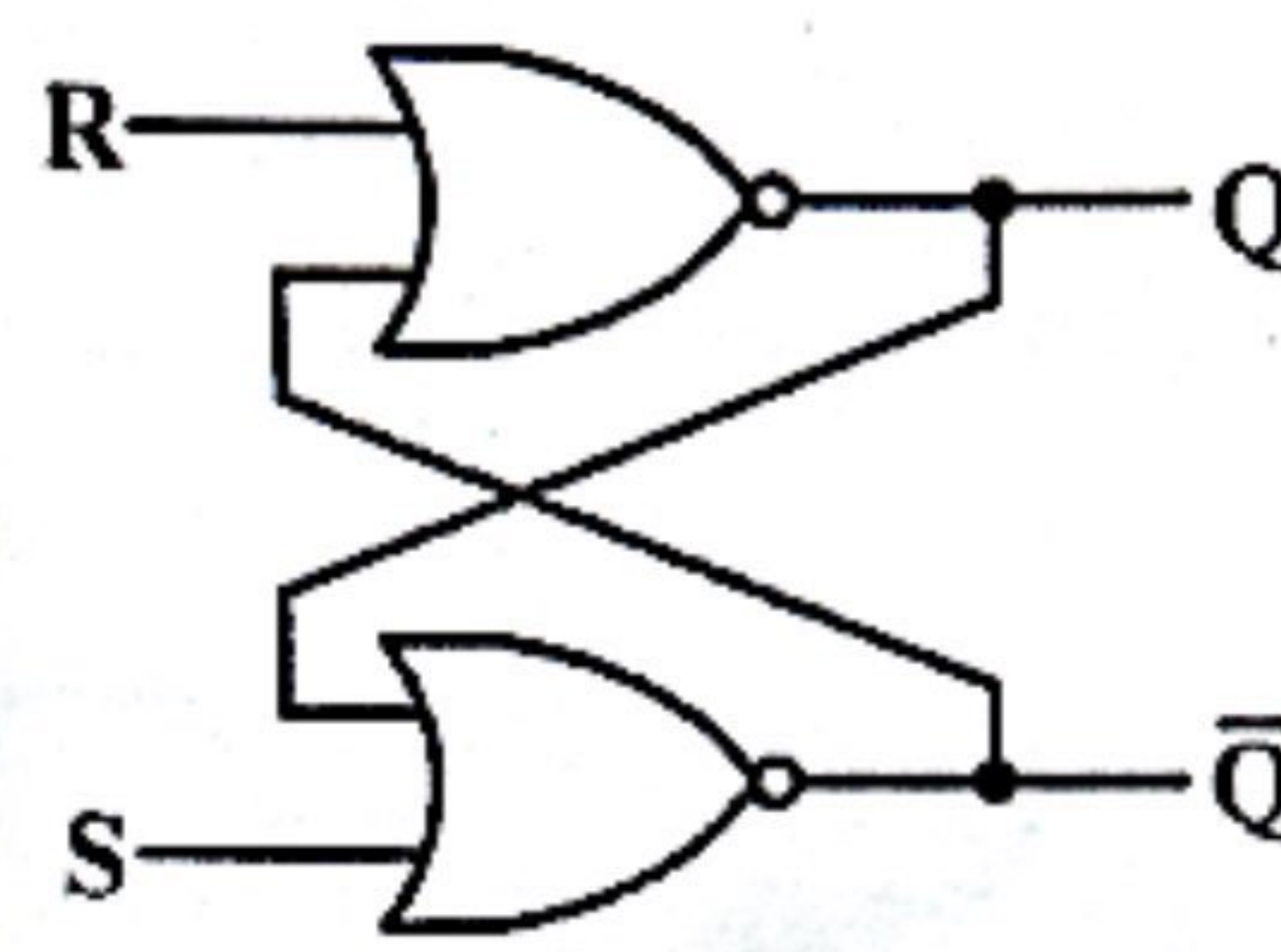


Fig. 3

- b) Sketch the circuit diagram to implement the following Boolean expression using static CMOS logic. [10] (CO2, PO2)

$$F = \overline{(AB + \bar{C})D}$$

- c) In Fig. 4, the I-V curves of nMOS and pMOS of a static CMOS inverter are shown for different values of input voltage (V_{in}) in a common coordinate set where the x-axis variable is the inverter output voltage (V_{out}) and y-axis variable is the current flowing from supply voltage towards the ground. With respect to Fig. 4, sketch the input (V_{in}) - output (V_{out}) characteristics curve of the CMOS inverter with proper labelling and highlight the different modes of operation of nMOS and pMOS along the curve. [5] (CO2, PO2)

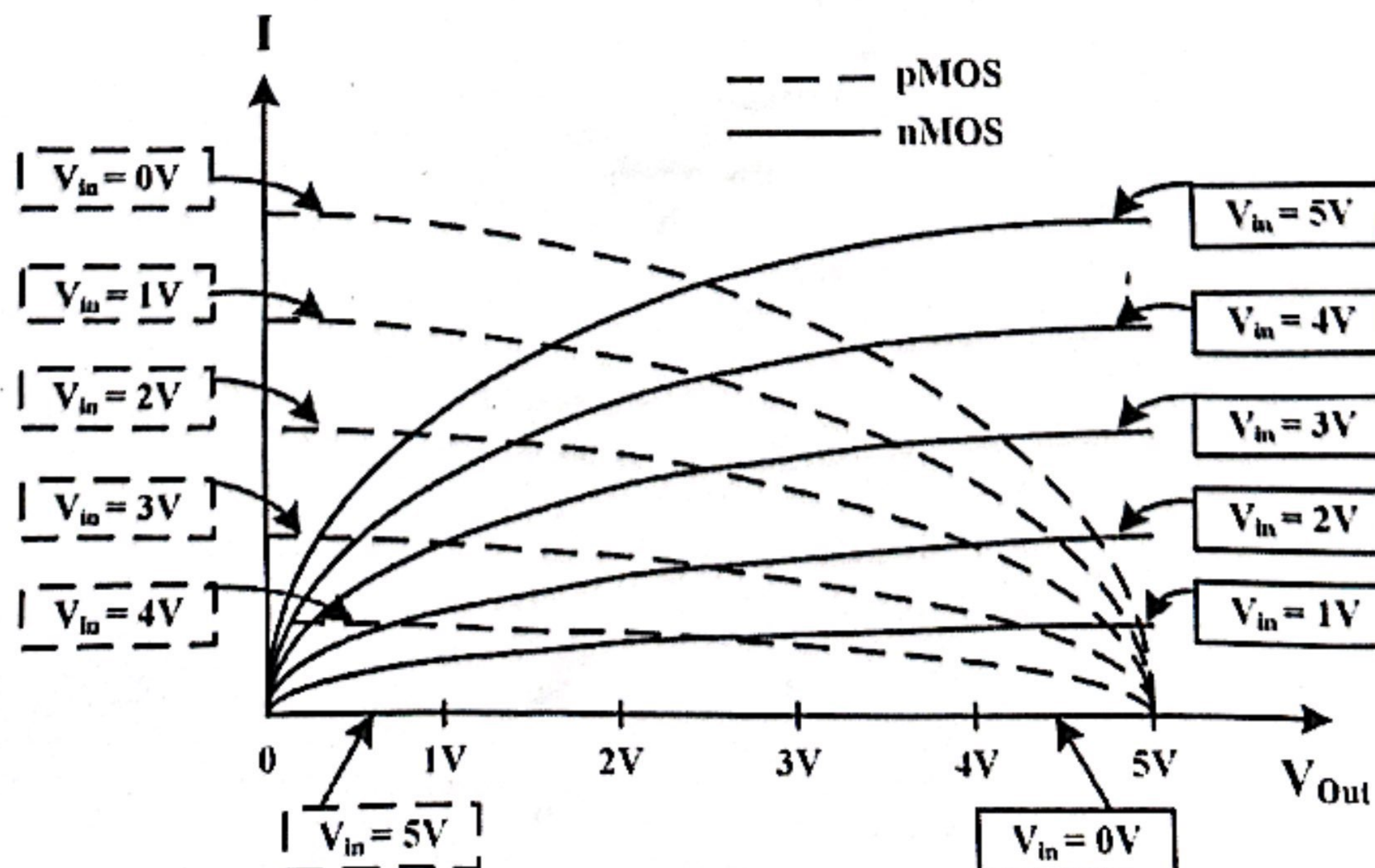


Fig. 4

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MID SEMESTER EXAMINATION

SUMMER SEMESTER, 2021-2022

DURATION: 1 HOUR 30 MINUTES

FULL MARKS: 100

CSE 4405: Data and Telecommunications

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) Umair and Abdullah are talking over landline (Traditional Phone). Landline uses permanent infrastructure made of wire. Identify different components Data Communication in the mentioned communication. How would this scenario change if they were talking over WhatsApp (a mobile app using traditional internet) video calls? 9
(CO1)
(PO1)
- b) How many levels of addresses are used in TCP/IP Protocol? What are the addresses and which OSI layer defining these addresses? Write down the significance or necessity of each level of address. In Figure 1, the Physical Address is represented with English letter and Logical Address is represented with numerical value. Example: Top-Left PC (sender) has physical address P and Logical Address 11. Here, sender with logical address 11 sends data to receiver with logical address 88. Draw the Datagrams and Frames with the '?' sign mentioning different addresses. 12
(CO1)
(PO1)

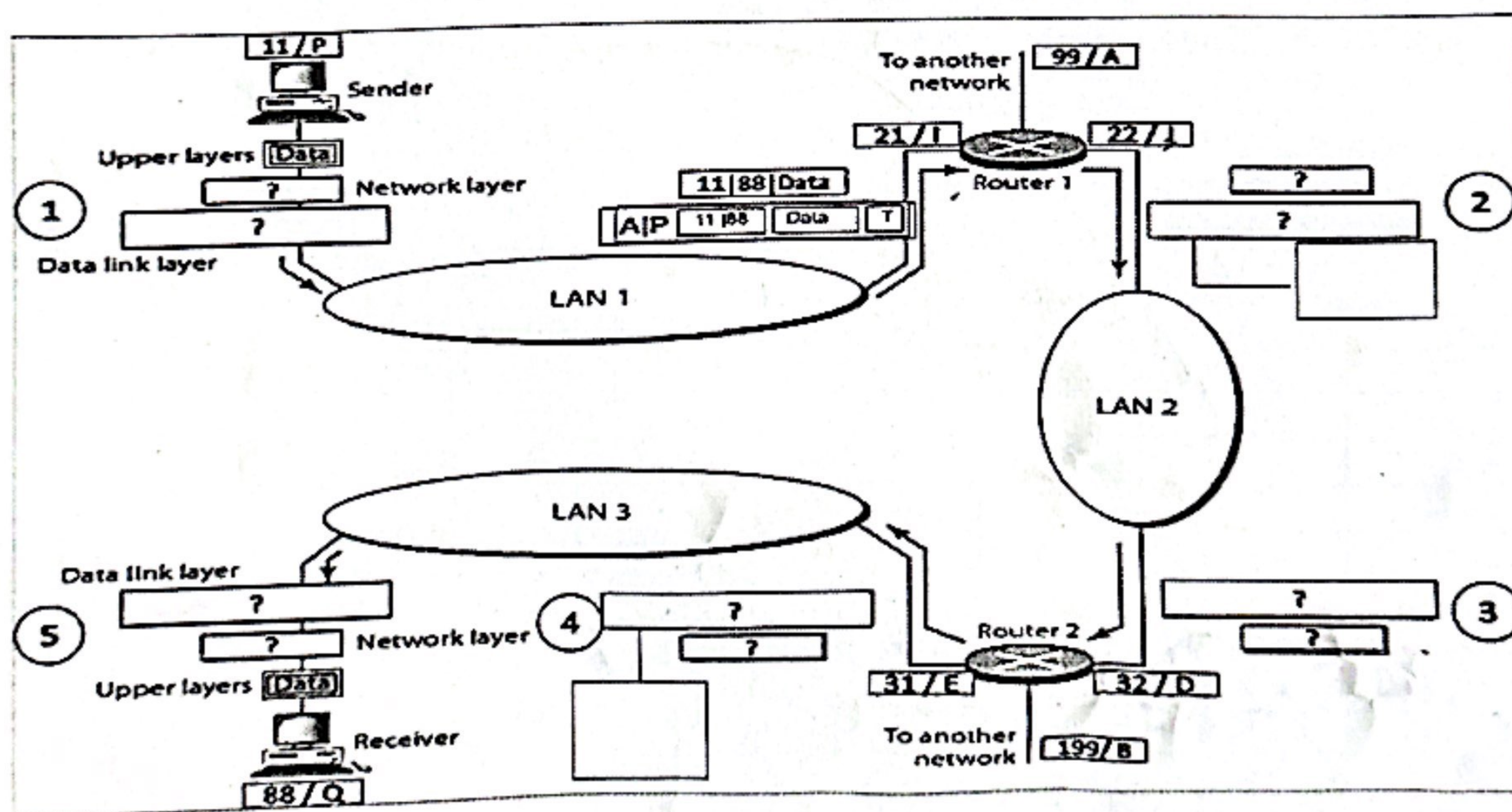


Figure 1: Figure for Question 1.b)

- c) How are OSI and ISO related to each other? How does OSI model differ from the TCP/IP protocol suite? Explain Process-to-Process delivery, Host-to-Host delivery, and Node-to-Node delivery in terms of the TCP/IP protocol suite. 12.33
(CO1)
(PO1)
2. a) What is Bandwidth-Delay product? Briefly explain different components of Latency (Delay). A frame of 1KB size is sent from source to destination. The transmission speed of the source is 1Mbps. A wired medium of CAT5 cable is used for this communication which has an average propagation speed of 2.4×10^8 m/s. There were no intermediate nodes and the distance between source and destination is 200 m. Calculate the total delay. 10
(CO2)
(PO2)
- b) Multiplexing and spreading are two ways of bandwidth utilization. How do they differ? What are the relative advantages and disadvantages of Statistical TDM over synchronous TDM? In a 10
(CO2)

synchronous TDM multiplexing technique, input data rates are 15, 20, 30, 40, and 40 kbps that need to be multiplexed and the output line has a capacity of 200 kbps. Draw the block diagram to show how disparity of data rates will be handled. Explain each of the techniques. (PO2)

OR

Give the taxonomy of digital-to-analog conversion techniques. Which of the techniques are most susceptible to noise? Justify your answer. Briefly explain the bandwidth requirements of different analog-to-analog conversion techniques. (CO2) (PO2) 10

c) 'In data communications, we commonly use periodic analog signal and non-periodic digital signal'. Justify the statement. Explain the concept of digital signal as a composite analog signal. Briefly explain different approaches to transmit a digital signal from one point to another point. (CO2) (PO2) 13.33

3. a) In NRZ encoding scheme, positive voltage defines bit 1 and zero voltage defines bit 0, In NRZ-L, level of the voltage determines the value of the bit. On the other hand, in NRZ-I, the change or lack of change in the level of the voltage determines the value of the bit. If no change the bit is 0, If there is change, the bit is 1. The idea of RZ (transition at the middle of a bit) and the idea of NRZ-L are combined into the Manchester scheme. Differential Manchester, on the other hand combines the idea of RZ and NRZ-I. There is always a transition at the middle of the bit, but the bit values are determined at the beginning of the bit. (CO2) (PO2) 9

Now explain the following pitfalls and explain in which scenarios of the input signals (Long string of 0 or long string of 1) these four schemes will face these pitfalls:

i. DC Component ii. Self-Synchronization iii. Baseline Wandering

b) 8B6T is a very interesting multilevel scheme used with 100BASE-4T cable where, a pattern of 8 bits is encoded as a pattern of 6 signal elements with three signal levels. The redundant signal elements provide synchronization, error correction, and DC balance. Convert the following hexadecimal string to 8B6T signal and provide the rationale of the above statements. (CO2) (PO2) 8

03 6A 69 20 49 2A

Draw the signal with notations. You can use multiple line if it does not fit in a single line. A partial table of 8B6T code is given in Table 1.

Table 1: Partial table of 8B6T code for Question 3.b)

Data	Code	Data	Code	Data	Code	Data	Code
00	-+00-+	20	-++-00	40	-00+0+	60	0++0-0
01	0-+-+0	21	+00+--	41	0-00++	61	+0+-00
02	0-+0-+	22	-+0-++	42	0-0+0+	62	+0+0-0
03	0-++0-	23	+-0-++	43	0-0++0	63	+0+00-
04	-+0+0-	24	+-0+00	44	-00++0	64	0++00-
05	+0--+0	25	-+0+00	45	00-0++	65	++0-00
06	+0-0-+	26	+00-00	46	00-+0+	66	++00-0
07	+0-+0-	27	-++++-	47	00-++0	67	++000-
08	-+00+-	28	0++-0-	48	00+000	68	0++++-
09	0-++-0	29	+0+0--	49	++-000	69	+0++++
0A	0-+0+-	2A	+0+-0-	4A	++-000	6A	+0+---
0B	0-+-0+	2B	+0+--0	4B	-++000	6B	+0+--+

c) MLT-3 is a differential coding scheme with more than two transition rules which maps one bit to one signal element (similar to NRZ-I). Explain the MLT-3 scheme with a suitable example and necessary state transition diagram. Justify the rationale of greater complexity (three level and complex transition rules). (CO2) (PO2) 10

d) State the Nyquist theorem for sampling rate used in PCM. Briefly explain quantization error of PCM. (CO2) (PO2) 6.34

OR

Assume that a voice channel occupies a bandwidth of 4-kHz. We need to combine three voice channels into a link with frequency spectrum from 50 to 65 kHz. Show the configuration using Block Diagram (both multiplexing and demultiplexing) using frequency domain. (CO2) (PO2) 6.34

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MID SEMESTER EXAMINATION
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SUMMER SEMESTER, 2021-2022
FULL MARKS: 75

Math 4441: Probability and Statistics

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. Consider a centralized wireless network where collision of packets are avoided by TDMA (Time-Division Multiple Access) scheduling. The success (or failure) of a packet delivery depends only on the wireless channel condition. If the wireless channel quality is good, the probability that a transmitted packet will be successfully delivered is 0.8. In contrast, if the wireless channel quality is poor, the probability that a transmitted packet will be successfully delivered is 0.6. Assume that the channel quality remains unchanged for a certain amount of time and after that it changes from the good state to the poor state and vice versa. Suppose the probability that the channel quality is good at a random time is 0.75 and the probability that the channel quality is poor is 0.25. Assume further that the probability of success (or failure) of a transmitted packet, at a certain channel quality (good or poor), is independent of the other packets transmitted at the same time instant (i.e., at the same channel quality).
 - a) Suppose the delivery status of five randomly selected packets is inspected at a certain time. Assume the channel quality is same during this time, i.e., the channel quality is either good or poor, but it does not change during this time. 10 + 7
(CO1)
(PO1)
 - i. Find the probability that four of the five transmitted packets are delivered successfully.
 - ii. Find the posterior probability that the channel condition was good at that time, given that four of the five transmitted packets were delivered successfully.
 - b) Suppose that the delivery status of one additional packet, which was delivered at the same time, is inspected again. 8 + 5
(CO1)
(PO1)
 - i. Find the probability that the additional inspected packet is delivered successfully using the conditional law of total probability.
 - ii. Find the new posterior probability that the channel condition was good at that time, given that four of the first five and the additional inspected packets were delivered successfully using the conditional Bayes theorem.
2. A computer executes two types of tasks, priority and non-priority, and operates in discrete time units (slots). A priority task arises with probability p at the beginning of each slot, independently of other slots, and requires one full slot to complete. A non-priority task is executed at a given slot only if no priority task is available. In this context, it may be important to know the probabilistic properties of the time intervals available for non-priority tasks.

With this in mind, let us call a slot busy if within this slot, the computer executes a priority task, and otherwise, let us call it idle. We call a sequence of idle (or busy) slots, surrounded by busy (or idle, respectively) slots, an idle period (or busy period, respectively).

 - a) Let T denote the time index of the first idle slot. Write down the PMF of T and justify it in no more than two sentences. 6
(CO1)
(PO1)
 - b) Suppose a non-priority task requires 5 slots to complete it. Assume no other non-priority task is present, and if any new non-priority task arrives, it will wait until this task is finished. Let S denote the number of slots required to complete the task. Write down the PMF of S and justify it in no more than two sentences. 7
(CO2)
(PO2)

c) Find the probability that there will be 10 busy slots before 5 idle slots.

9

(CO2)

(PO2)

3. Suppose that a professor will take a lab viva of five students one after another. The number of minutes required by any particular student to complete the viva has the exponential distribution for which the mean is 30. Suppose that the examination begins at 9:00 a.m.

a) Determine the probability that the first student will complete the examination before 9:40 a.m.

7

(CO1)

(PO1)

b) Determine the probability that it will be at least 11:00 a.m. to complete the viva of the first three students.

10

(CO2)

(PO2)

c) Find the expected time required to complete the viva of all five students.

6

(CO1)

(PO1)